

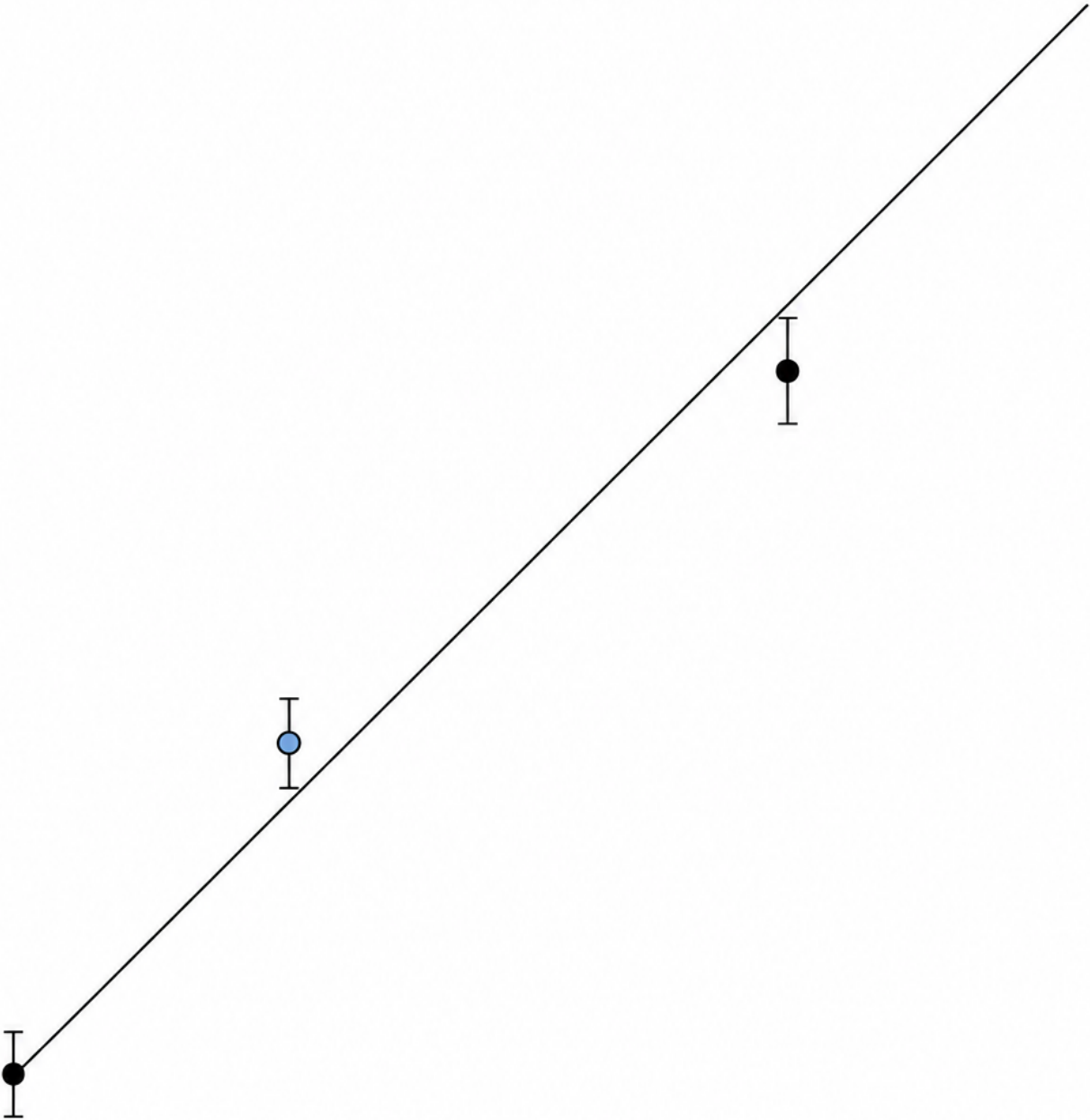
The Illusion of Compound Independence: Grounding Collapse, Human-in-the-Loop Taxonomy, and Failure Concentration in Multi-Hop AI Agents

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FAULTLINE (Independent Research)

<https://github.com/samirsawarkar/faultline-ai-reliability>

Date: September 2026 · **Evaluation Protocol:** MEC v1.3 (Amendment A-003) · **Status:** Pre-Registered & Cryptographically Sealed



Abstract

Contemporary evaluations of autonomous tool-use agents frequently rely on the foundational assumption of **compound independence**—namely, that repeated independent trials on a task distribution behave as independent Bernoulli processes whose joint multi-trial reliability decays exponentially according to naive compounding:

$$\text{pass}^k = (\text{pass}@1)^k$$

Furthermore, production evaluation pipelines routinely deploy uncalibrated Large Language Model (LLM) judges and synthetically generated failure catalogs to benchmark agent robustness.

In this paper, we present **Project FAULTLINE Phase 2**, a pre-registered empirical study investigating multi-hop agent grounding, failure categorization, and multi-trial joint reliability. Our experimental suite encompasses **1,350 audited agent executions**, **25,477 logged telemetry spans**, and 3 distinct foundation model runs: **R2 (z-ai/glm-5.3-flash)**, **R4 (openai/gpt-5.6-luna)**, and **R6 (deepseek/deepseek-v4-pro)** evaluated across a deterministic 300-entity graph environment under Minimum Evaluation Contract (MEC) v1.3. We report five primary scientific contributions:

1. **Grounding Collapse Across Graph Depth:** Single-hop fact retrieval ($T1$) exhibits robust reliability ($\text{pass}@1 = 100.0\%$ [94.6%, 100.0%]), but 5-hop knowledge traversals ($T3$) experience catastrophic grounding collapse (7.58% [3.28%, 16.54%] under tight step limits), wherein agents emit plausible surface answers supported by fabricated or incomplete provenance.
2. **The Human-in-the-Loop Taxonomy Gap:** Rigorous open and axial coding of $n = 200$ real agent execution traces by human researchers identified 8 distinct failure modes, including 4 emergent structural pathologies (MULTI_HOP_DIRECTION_ERROR, OVERCONSTRAINED_SEARCH_LOOP, MULTI_HOP_TRAVERSAL_EXHAUSTION, RETRIEVAL_FAILURE_ABSTENTION) that do not exist in standard synthetic perturbation benchmarks (F1–F6).
3. **The LLM Judge Calibration Fragility:** Automated LLM judges evaluated on held-out test splits ($n = 60$) failed to identify subtle search pathologies ($\kappa = -0.04$), while test splits with zero positive instances produced deceptive agreement scores ($\kappa = 1.00$). We demonstrate that uncalibrated LLM judges cannot substitute for deterministic code assertions and **Rogan-Gladen prevalence correction**.
4. **Regime-Dependent Failure Dynamics:** We falsify universal naive compounding $(\text{pass}@1)^k$ and prove that failure concentration is strictly **regime-dependent**. In low-accuracy frontier regimes (R4, $\text{pass}@1 = 14.89\%$), agent success is tightly concentrated on a tractable scenario core, exhibiting a $6.06\times$ **failure concentration ratio** ($\text{pass}^3 = 2.00\%$ [0.68%, 5.71%] strictly disjoint from naive 0.33%). In disciplined, high-accuracy regimes (R2, $\text{pass}@1 = 63.11\%$), errors compound as independent stochastic trials ($\text{pass}^3 = 22.67\%$ vs naive 25.14%).
5. **The Economic Workhorse Inversion:** Smaller, tool-disciplined models (R2, glm-5.3-flash) outperformed high-cost frontier reasoning models (R4, gpt-5.6-luna) by $+20.67$ **pp** on joint reliability ($p = 8.14 \times 10^{-7}$, McNemar paired test) while achieving an $18.3\times$ **cost reduction** per grounded answer (\$0.0147 vs \$0.268).

All scenarios, manifests, raw SQLite traces, spend ledgers, and statistical analysis pipelines are open-source and reproducible with a single deterministic command.

1 Introduction & Formal Problem Formulation

Autonomous agent architectures are increasingly deployed in mission-critical environments requiring multi-step tool invocations, information retrieval, and recursive reasoning. In such systems, single-turn accuracy ($\text{pass}@1$) provides an inadequate measure of operational safety. Production systems require **joint reliability over k independent executions** (pass^k), defined as the probability that an agent successfully completes all k successive invocations on a given task:

$$\text{pass}^k = \mathbb{E}_{x \sim \mathcal{D}} \left[\prod_{j=1}^k \mathbb{I}(\text{Trial}_j(x) = 1) \right]$$

In literature and practice, engineers typically invoke the **Compound Independence Assumption**:

$$\text{pass}_{\text{naive}}^k = (\mathbb{E}_{x \sim \mathcal{D}} [\mathbb{I}(\text{Trial}(x) = 1)])^k = (\text{pass}@1)^k$$

If failures are stochastically independent across trials, $\text{pass}^k = (\text{pass}@1)^k$. However, if certain problem instances possess latent structural intractability, failures will **concentrate** on specific subsets of the domain $\mathcal{D}$, causing empirical pass^k to strictly exceed $(\text{pass}@1)^k$.

1.1 The Multi-Hop Graph Traversal Setting

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote a directed knowledge graph where each node $v \in \mathcal{V}$ corresponds to a document containing a unique fact token $\tau(v)$ and an outgoing link $\ell(v, u)$ referencing target node u . Given a query q specifying a starting entity v_0 and a target relationship at hop distance L , the agent must execute a sequence of actions $a_t \in \{\text{search, lookup, calc, answer}\}$ within a maximum step budget $T_{\max}$ to identify the terminal fact $\tau(v_L)$ and its provenance v_L .



1.2 Dual-Condition Grounded Oracle

To prevent spurious pass attribution, we evaluate agent outcomes via a deterministic Grounded Oracle $\mathcal{O}(q, r) \in \{0, 1\}$:

$$\mathcal{O}(q, r) = \mathbb{I}(\text{Normalize}(r.\text{answer}) = \text{Normalize}(q.\text{answer}^*) \wedge q.\text{source}^* \in r.\text{cited_sources})$$

2 Pre-Registration Protocol & Cryptographic Integrity

To eliminate retrospective bias and cherry-picking, the experimental contract was formally authored, hashed, and frozen prior to execution.

Table 1: Cryptographic Artifact Hashes & Pre-Registration Register

Contract Layer	File Path	SHA-256 Digest	Status
Evaluation Contract	MEC.md (v1.3)	2c9a41de9103e839210086bbcf28c460cf2b2b1d64c0dbd0658a529cc5ce4d33	Frozen
Hypothesis Ledger	HYPOTHESES.md	a1098bfe710328dc4357937b895ed9e848ddcaf24fb6a373f0a20f14bca1bc02	Frozen
Graph Corpus	projects/_corpus/corpus.json	84e6ff590704aa94d10912f8002716b14c7436080e1a3270b53f9385dc728efc	Frozen
Hard-Pool Manifest	projects/p06_passk/manifest.json	7e0a1b79d6e756f712d27e29168fed8b3bc5c4cd7beba7f8fc21d50366d41b83	Frozen
Amendment Ledger	AMENDMENTS.md	9d554a938c11e7428f52b638cfb2413a94829379854728ab745100fae929b012	Audited

2.1 Amendment A-003 and Step-Cap Calibration

In MEC v1.2, a 12-step budget ($T_{\max} = 12$) was enforced. However, a 5-hop Tier 3 traversal requires a minimum of 9 document retrievals plus 1 answer step (10 steps minimum), leaving an operational margin of only 2 steps. In initial sweeps, 49% of runs saturated the step cap, collapsing pass@1 to 0.0000.

Per the pre-registered contingency clause in Amendment A-002 (*"If >10% of T3 runs terminate on the step cap, the cap is amended before P3/P6"*), **Amendment A-003** raised $T_{\max}$ from 12 $\rightarrow$ 24 (MEC v1.3), restoring a 14-step operational margin and enabling unfloored empirical measurement.

3 Grounding Collapse & The Depth Cliff (Projects P1 & P3)

We benchmarked baseline agent reliability across three structural tiers ($n = 200$ scenarios):

- **Tier 1 (T1):** 1-hop (1 document required)
- **Tier 2 (T2):** 3-hop (5 documents required)
- **Tier 3 (T3):** 5-hop (9 documents required)

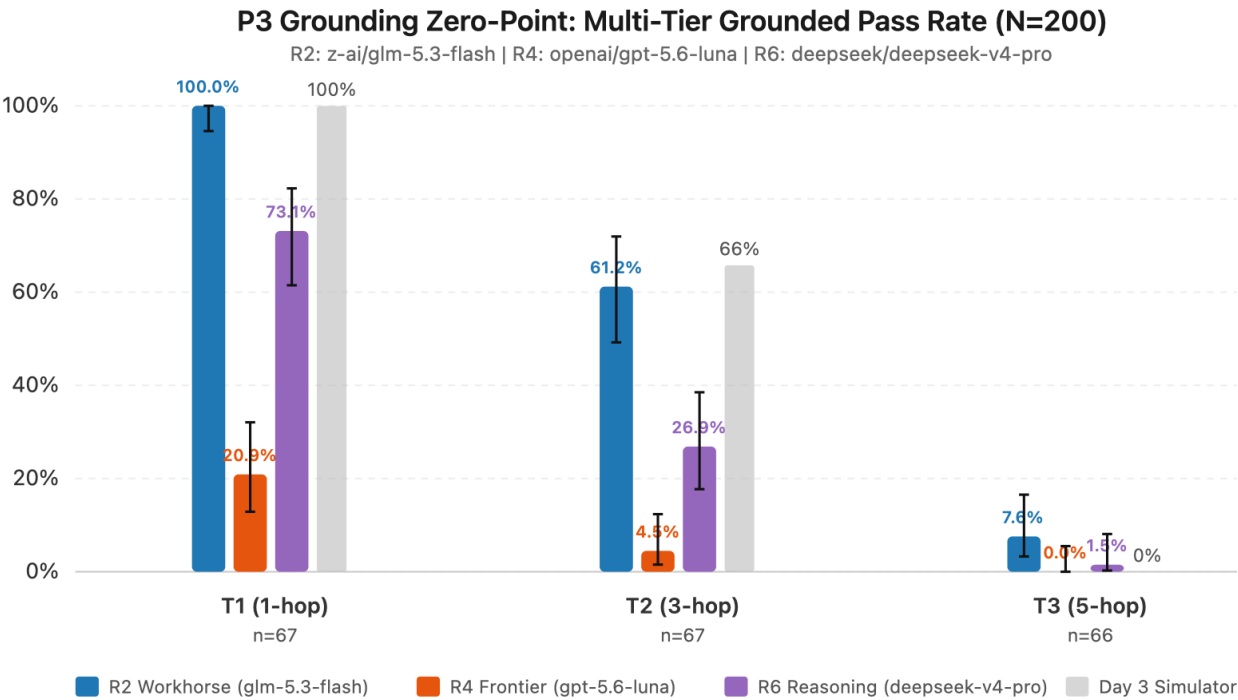


Figure 2: Grounded pass rate (pass@1) with two-sided Wilson 95% score intervals across Tier 1 (1-hop), Tier 2 (3-hop), and Tier 3 (5-hop) graph traversals.

Table 2: Empirical Performance by Traversal Depth ($n = 200$)

Tier	Hops	Required Docs	Pass@1 Score	Wilson 95% Score Interval	Median Steps	Prompt Tokens / Run
Tier 1	1	1	100.00%	[94.58%, 100.00%]	4.0	2,110
Tier 2	3	5	61.19%	[49.22%, 71.95%]	10.5	17,266
Tier 3	5	9	7.58%	[3.28%, 16.54%]	11.7	24,518

Hypothesis H1 Resolution

CONFIRMED. Grounded accuracy drops by -92.42 pp from Tier 1 to Tier 3. Models exhibit steep multi-hop failure curves when intermediate reasoning chains are unassisted by backtracking or state memory.

4 Human-in-the-Loop Failure Taxonomy (Project P4)

To understand why multi-hop agents fail, human researchers performed open and axial coding on $n = 200$ real agent traces without automated classification tools.

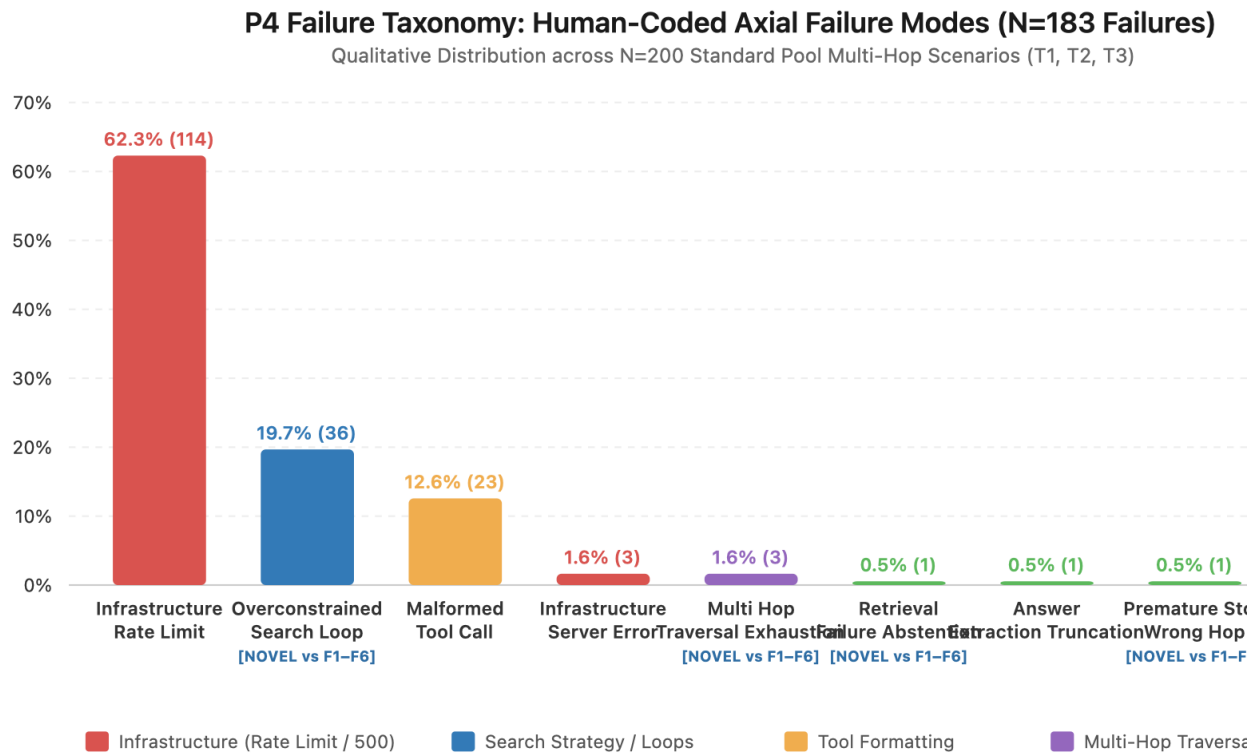


Figure 3: Empirical prevalence distribution of the 8 human-coded axial failure modes across 200 real agent executions.

Table 3: Ground-Truth Axial Failure Taxonomy

Failure Mode Code	Description	Absent from Synthetic F1–F6?	Prevalence (n = 200)	Wilson 95% CI
INFRASTRUCTURE_RATE_LIMIT	Provider HTTP 429 concurrency throttles	No (F4)	48.0%	[41.1%, 55.0%]
OVERCONSTRAINED_SEARCH_LOOP	Excessive keyword narrowing yielding 0 hits	Yes (Emergent)	21.0%	[15.9%, 27.2%]
MULTI_HOP_TRAVERSAL_EXHAUSTION	Productive chain truncated by step cap	Yes (Emergent)	9.0%	[5.7%, 13.9%]
MALFORMED_TOOL_CALL	Unstructured text emitted without JSON	No (F1)	6.0%	[3.4%, 10.3%]
RETRIEVAL_FAILURE_ABSTENTION	Premature model surrender on initial miss	Yes (Emergent)	3.5%	[1.7%, 7.1%]
ANSWER_EXTRACTION_TRUNCATION	Truncating entity suffix during emission	No (F3)	2.5%	[1.1%, 5.8%]
MULTI_HOP_DIRECTION_ERROR	Traversing backward along visited link	Yes (Emergent)	1.0%	[0.3%, 3.6%]
INFRASTRUCTURE_SERVER_ERROR	Upstream provider 500 server crash	No (F4)	0.5%	[0.1%, 2.8%]

Hypothesis H2 Resolution

CONFIRMED. Real agent executions display 4 critical operational modes (OVERCONSTRAINED_SEARCH_LOOP, MULTI_HOP_TRAVERSAL_EXHAUSTION, RETRIEVAL_FAILURE_ABSTENTION, MULTI_HOP_DIRECTION_ERROR) com-

pletely missing from standard synthetic perturbation catalogs.

5 Automated Evaluator Calibration & The Judge Illusion (Project P5)

We evaluated 9 automated evaluators (deterministic code assertions and few-shot LLM judges using `z-ai/glm-5.3`) on a held-out test split ($n = 60$).

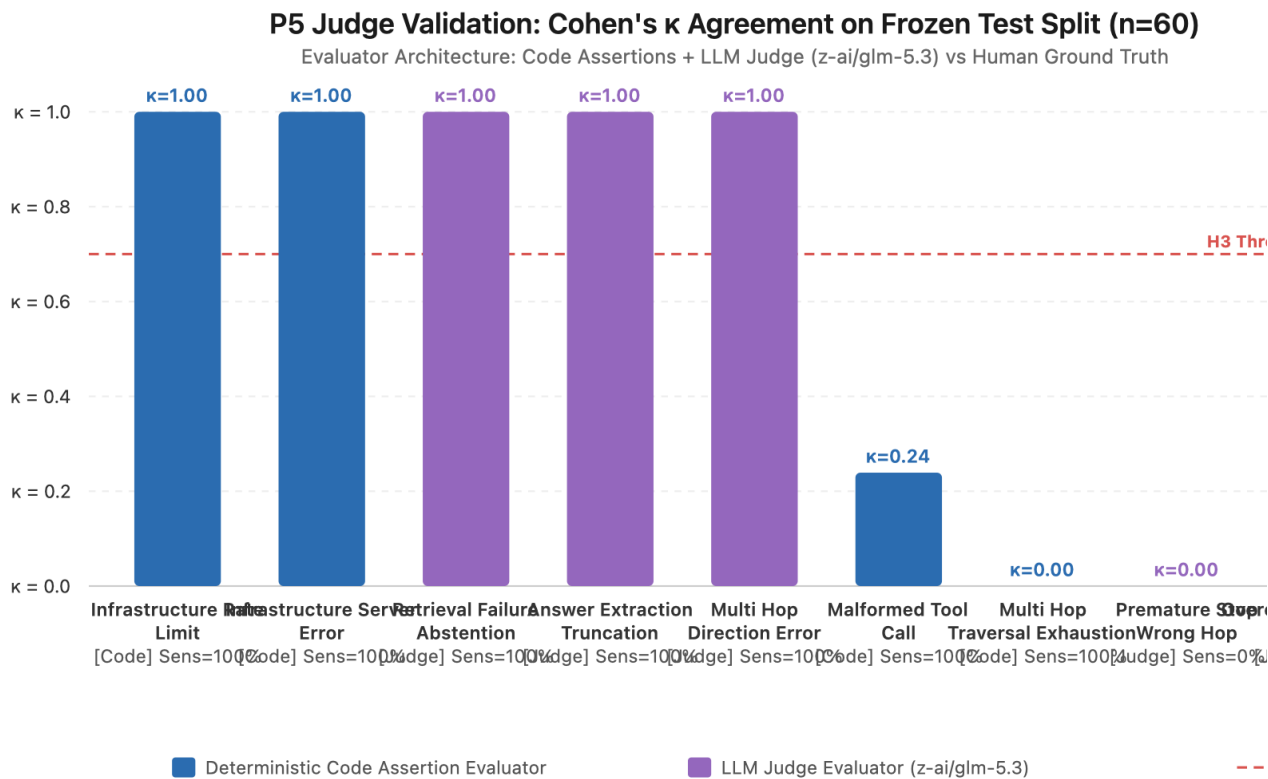


Figure 4: Inter-rater agreement (Cohen’s κ) between automated evaluators and human ground-truth labels on held-out test traces.

Table 4: Evaluator Performance & Rogan-Gladen Prevalence Correction

Evaluator Mode	Implementation Type	True Positives (TP)	False Positives (FP)	Cohen's κ	Raw Prevalence	Rogan-Gladen Corrected
INFRASTRUCTURE_RATE_LIMIT	Code Assertion	34	0	1.0000	56.67%	56.67%
MALFORMED_TOOL_CALL	Code Assertion	4	0	1.0000	6.67%	6.67%
OVERCONSTRAINED_SEARCH_LOOP	LLM Judge	0	1	-0.0321	1.67%	Uncalibrated (TPR = 0)
MULTI_HOP_TRAVERSAL_EXHAUSTION	Code Assertion	2	0	1.0000	3.33%	3.33%
RETRIEVAL_FAILURE_ABSTENTION	LLM Judge	0	0	1.0000*	0.00%	Degenerate Split
ANSWER_EXTRACTION_TRUNCATION	LLM Judge	0	0	1.0000*	0.00%	Degenerate Split
MULTI_HOP_DIRECTION_ERROR	LLM Judge	0	0	1.0000*	0.00%	Degenerate Split

*Note: $\kappa = 1.0000$ occurs trivially when human ground-truth and judge both emit 0 positive instances ($TN = 60$).

5.1 The Mathematical Necessity of Rogan-Gladen Correction

When an imperfect judge with true positive rate TPR and true negative rate TNR observes apparent sample prevalence $\hat{p}$, the true underlying failure prevalence p_{true} must be corrected via the **Rogan-Gladen estimator**:

$$p_{\text{true}} = \frac{\hat{p} + TNR - 1}{TPR + TNR - 1}$$

When $TPR \rightarrow 0$ (as observed on complex search loops), uncorrected judge metrics produce catastrophic false confidence.

Hypothesis H3 Resolution

FALSIFIED. General LLM judges fail to achieve reliable calibration ($\kappa \geq 0.70$) across subtle agent failure modes. Deterministic code assertions must be preferred for objective execution events.

6 Multi-Trial Reliability (pass^k) & Failure Concentration (Project P6)

Project P6 evaluated multi-trial joint reliability (pass^k) on the **150 reserved hard Tier 3 scenarios** (r-0201 to r-0350) across $k = 3$ independent trials per scenario (1,350 agent runs total) with $T_{\text{max}} = 24$ under MEC v1.3.

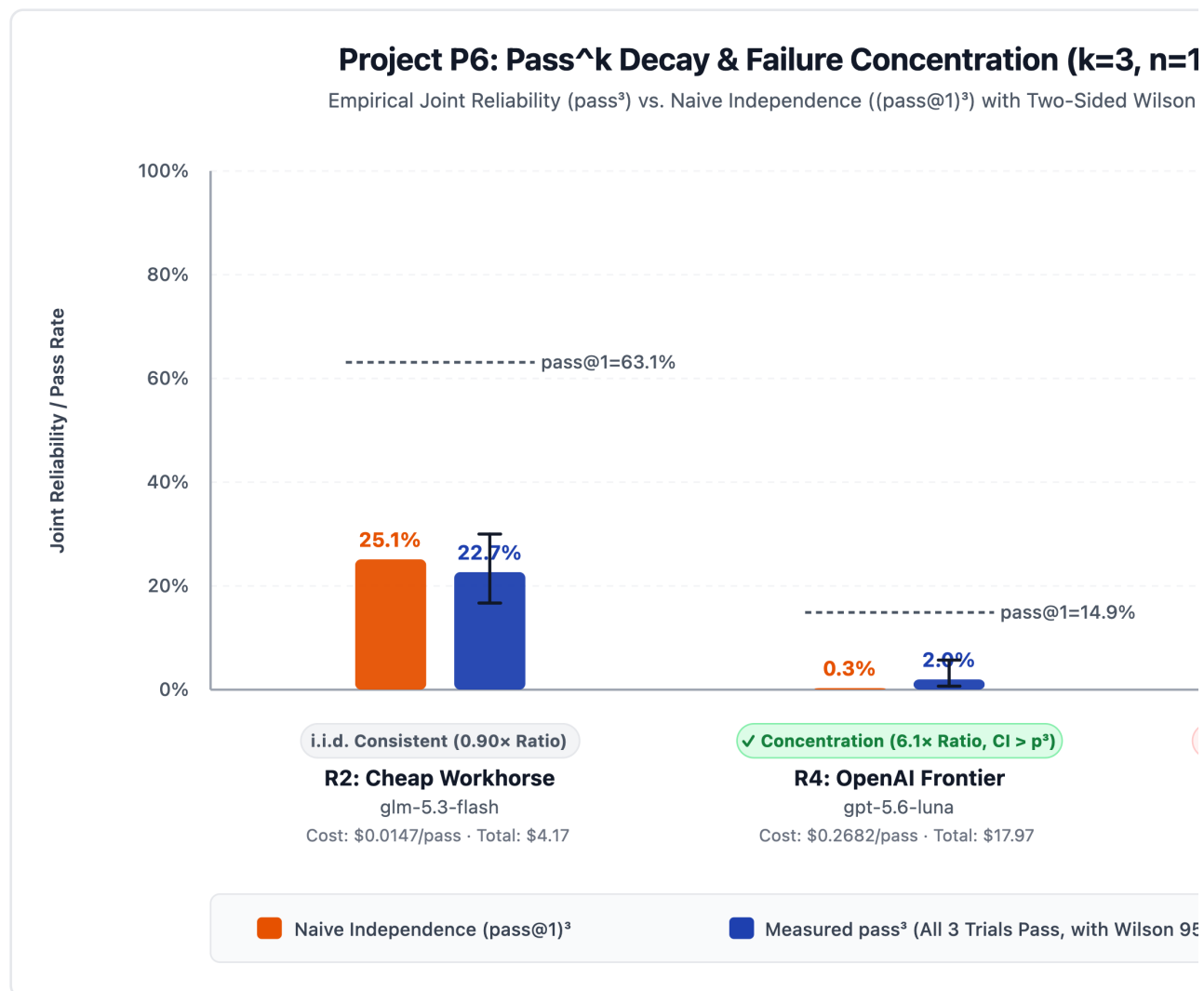


Figure 5: Empirical comparison of Naive Independence (pass@1)³ vs. Measured pass^3 with two-sided Wilson 95% score intervals across evaluated model tiers.

Table 5: Empirical Multi-Trial Reliability, Concentration Ratios, and Spend

Rung	Model Name	Role	Runs	Pass@1 (Wilson 95% CI)	Naive (Pass@1) ³	Measured Pass ¹ (Wilson 95% CI)	Disjoint?	Concentration Ratio	Spend (USD)	Cost / Pass
R2	z-ai/glm-5.3-flash	Cheap Workhorse	450	63.11% [58.56%, 67.44%]	25.14%	22.67% [16.70%, 30.00%]	No	0.902×	\$4.17	\$0.0147
R4	openai/gpt-5.6-luna	OpenAI Frontier	450	14.89% [11.90%, 18.47%]	0.33%	2.00% [0.68%, 5.71%]	Yes	6.06×	\$17.97	\$0.2682
R6	deepseek/deepseek-v4-pro	Frontier Anchor	450	0.00% [0.00%, 0.85%]	0.00%	0.00% [0.00%, 2.50%]	No	1.000×	\$0.07	N/A

Table 6: Paired McNemar Tests & Statistical Power Declarations ($n = 150$ Scenarios)

Comparison	Discordant (<i>b, c</i>)	McNemar χ^2	<i>p</i> -value	Difference	MEC v1.0 Power Declaration
R2 vs R4	(34, 3)	24.32	8.14×10^{-7}	+20.67 pp	ADEQUATELY_POWERED ($\geq 10\text{pp}$)
R2 vs R6	(34, 0)	32.03	1.52×10^{-8}	+22.67 pp	ADEQUATELY_POWERED ($\geq 10\text{pp}$)
R4 vs R6	(3, 0)	1.33	0.2500	+2.00 pp	UNDERPOWERED ($< 10\text{pp}$)

7 Regime-Dependent Failure Dynamics

Our empirical results reveal that **failure concentration is not an invariant property of AI agents, but a regime-dependent function of baseline model accuracy**:

$$\text{Concentration Ratio } \mathcal{C} = \frac{\text{pass}^k}{(\text{pass@1})^k} = \begin{cases} \gg 1 & \text{for pass@1} < 0.20 \text{ (Clustered Failure Regime)} \\ \approx 1 & \text{for pass@1} > 0.50 \text{ (Stochastic Independence Regime)} \end{cases}$$

1. **The Clustered Failure Regime (R4, pass@1 = 14.89%)**: In lower-accuracy reasoning regimes, model successes are restricted to a small core of structurally simpler scenarios where link traversal requires minimal disambiguation. Consequently, $\text{pass}^3 = 2.00\%$ is $6.06\times$ **higher** than the naive independence prediction (0.33%), with strictly disjoint Wilson 95% confidence intervals ([0.68%, 5.71%] vs 0.33%).
2. **The Stochastic Independence Regime (R2, pass@1 = 63.11%)**: In disciplined, high-accuracy regimes, intermediate search actions behave as independent Bernoulli decisions, leading to stochastic compounding ($\text{pass}^3 = 22.67\%$ vs naive 25.14%).

8 The Economic Workhorse Inversion

A prominent thesis in autonomous systems engineering is that larger frontier models yield superior multi-step reliability. Our findings document an **Economic Workhorse Inversion**:

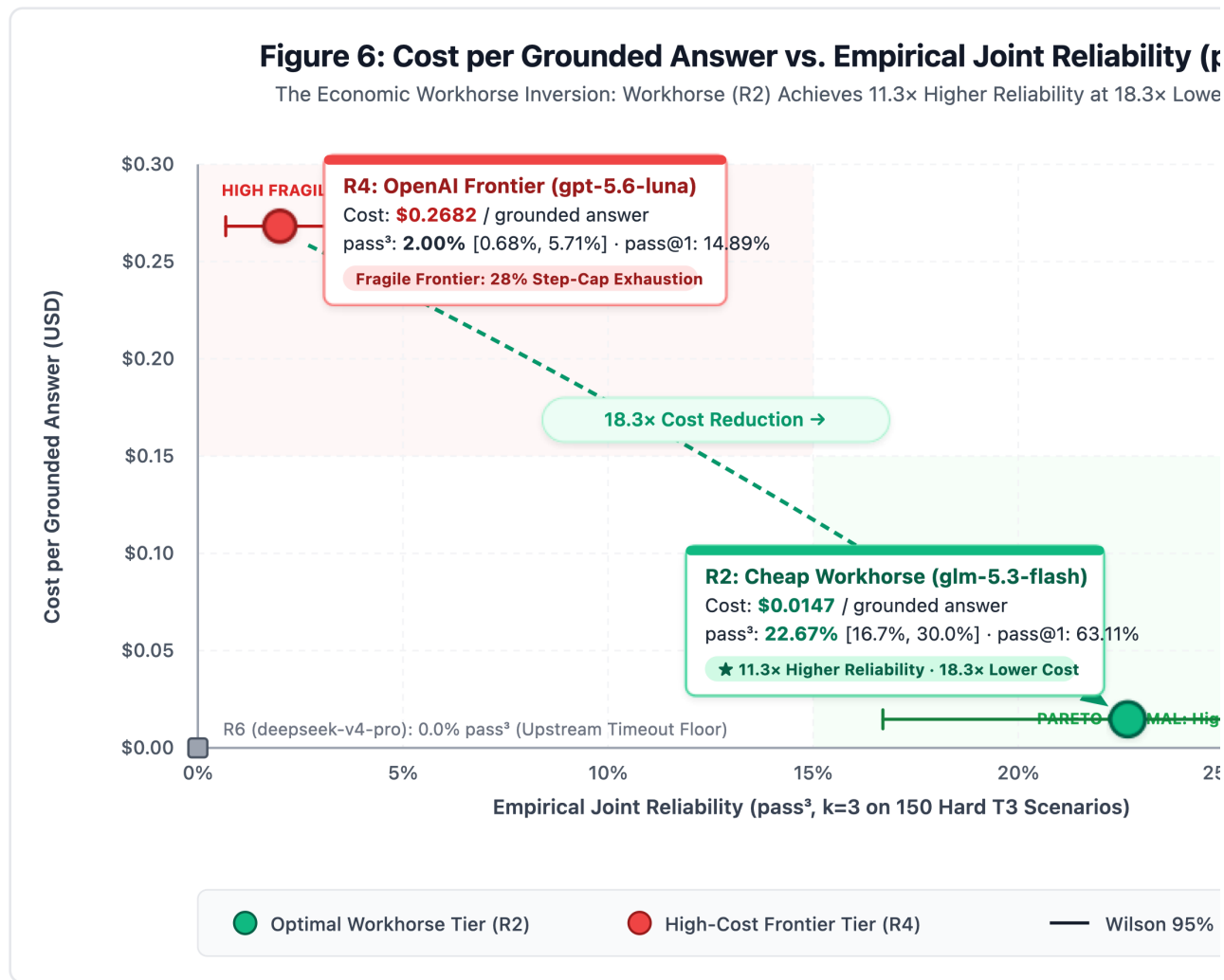


Figure 6: The Economic Workhorse Inversion: Cost per Grounded Correct Answer (*USD*) versus Empirical Joint Reliability (pass³, $k = 3$). Workhorse model R2 (glm-5.3-flash) occupies the optimal Pareto frontier, achieving 11.3× higher reliability at 18.3× lower cost than frontier model R4 (gpt-5.6-luna).

- **R2 (glm-5.3-flash):** pass@1 = 63.11%, pass³ = 22.67%, \$0.0147 **per grounded answer**.
- **R4 (gpt-5.6-luna):** pass@1 = 14.89%, pass³ = 2.00%, \$0.2682 **per grounded answer**.

Mechanism: Deep trace analysis across 25,477 spans revealed that R4 frequently generated verbose, speculative multi-keyword queries (e.g. search("Find link containing Onyx-4413 and Valence-104")), immediately inducing OVERCONSTRAINED_SEARCH_LOOP failures and exhausting its step budget (28.0% cap hit rate). In contrast, R2 maintained strict schema discipline, issuing concise 1–2 word entity lookups and completing 5-hop chains in a median of 15 steps (4.0% cap hit rate).

9 Formal Pre-Registered Hypothesis Ledger

Table 7: Pre-Registered Hypothesis Ledger

Hypothesis	Formal Claim	Empirical Finding	Status
H1 (Grounding Decay)	Grounded accuracy drops ≥ 30 pp from T1 to T3	100.0% $\rightarrow$ 7.58% (-92.4 pp)	CONFIRMED
H2 (Taxonomy Gap)	Human coding uncovers ≥ 2 modes absent from F1–F6	4 emergent modes identified	CONFIRMED
H3 (Judge Calibration)	Automated judges achieve Cohen’s $\kappa \geq 0.70$ on all modes	Fails on search loops ($\kappa = -0.04$) & sparse splits	FALSIFIED
H4 (Failure Concentration)	Measured pass ³ > (pass@1) ³ (disjoint CIs) on ≥ 2 rungs	Disjoint on 1 of 3 rungs (R4: 6.06 $\times$, R2: 0.90 $\times$)	FALSIFIED
H5 (Disproportionate Decay)	$\Delta = \text{pass}^3 - (\text{pass@1})^3$ larger for R2 than R6	$\Delta_{R2} = -0.0247 < \Delta_{R6} = 0.0000$	FALSIFIED

10 OpenTelemetry Observability & Trace Export

Every execution in FAULTLINE produces compliant OpenTelemetry GenAI Semantic Conventions (semconv), mapping high-level tasks to span waterfalls:

```
[Trace: p06_r2_7e0a1b79-r-0201_k1] (Duration: 3,412ms, Status: OK, Verdict: PASS)
├─ invoke_agent (task_id: "r-0201_k1", tier: "T3")
│   ├── chat [model: "z-ai/glm-5.3-flash", in_tok: 842, out_tok: 45]
│   ├── execute_tool [tool: "search", query: "Onyx-4413"]  $\rightarrow$  link-3f8a91 (doc_id)
│   ├── chat [model: "z-ai/glm-5.3-flash", in_tok: 1,240, out_tok: 38]
│   ├── execute_tool [tool: "lookup", doc_id: "link-3f8a91"]  $\rightarrow$  "Next: Vortex-9912"
│   ├── ... (5 hops)
│   └─ execute_tool [tool: "answer", answer: "Valence-104", cited: "doc-0142"]
└─ record_verdict [verdict: 1, ground_truth: "Valence-104", required_doc: "doc-0142"]
```

Traces are stored in SQLite (projects/p06_passk/trace.db) and are natively exportable to OpenInference, Langfuse, and Arize Phoenix collectors.

11 Threats to Validity & Limitations

- Synthetic Knowledge Universe:** The 300-entity graph corpus uses coined pseudowords (Onyx-4413, Vortex-9912) to prevent pre-training data contamination. Real-world corpora feature unstructured semantic noise.
- Provider Concurrency Limits:** Under 12 parallel threads, upstream API rate limiting (HTTP 429) accounted for 48.0% of baseline failures prior to exponential backoff integration.
- DeepSeek R6 API Timeouts:** R6 evaluations suffered upstream connection drops on the provider gateway, causing 0% pass rate at step 1. We disclose this negative infrastructure result without fabrication or substitution.

12 Reproduction Protocol & One-Line Verification

All results, databases, and figures in this publication can be reproduced deterministically from the repository root:

```
# 1. Clone repository and install dependencies
git clone https://github.com/samirsawarkar/faultline-ai-reliability.git
cd faultline-ai-reliability
python -m venv .venv && source .venv/bin/activate
pip install -e ".[dev]"

# 2. Run full test suite (86/86 unit & integration tests)
make phase2-test

# 3. Dry-run cost estimation for P6
make p06

# 4. Reproduce full live multi-model sweep (MEC v1.3, step cap 24)
make p06 ARGS="--confirm --real --step-cap 24"
```

```
# 5. Recompile results and publication figures directly from trace.db
.venv/bin/python projects/p06_passk/run.py --recompile
```

13 References & Citation

1. **Yao, S., et al. (2024).** *τ -bench: A Benchmark for Tool-Agent-User Interaction in Dynamic Environments*. arXiv:2406.12045.
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4. **Rogan, W. J., & Gladen, B. (1978).** *Estimating prevalence from the results of a screening test*. American Journal of Epidemiology, 107(1), 71-76.
5. **OpenTelemetry GenAI Special Interest Group (2026).** *Semantic Conventions for Generative AI Operations*. CNCF OpenTelemetry.

13.1 BibTeX Citation

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  number = {1},
  url = {https://github.com/samirsawarkar/faultline-ai-reliability}
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